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Electromagnetic and mechanical properties of carbonyl iron powders-PLA composites fabricated by fused deposition modeling

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Abstract

3D printing is an emerging additive manufacturing method for effectively fabricating materials with complex structures. Fused deposition modeling (FDM) is one of the most popular 3D printing techniques, usually used for making thermoplastic parts. However, this method has been rarely considered for designing and fabricating radar wave absorbing structures, which can be very helpful to achieving intricate structures with good mechanical and wave absorbing properties. To verify the validity of this method, carbonyl iron particles filled thermoplastics composites are fabricated by FDM in this study. The morphological, magnetic, electromagnetic and mechanical properties of the composites are investigated by scanning electron microscopy, vibrating sample magnetometer, broad band dielectric spectrometer and universal testing machine. The composites exhibit the hysteretic loops of typical soft ferromagnetism with relatively low remanence, coercive force, and hysteresis losses. The saturation magnetization per unit mass of iron powders of the composites does not have a linear relation with filler contents. The obtained composites materials possess slightly improved tensile strength and significantly improved total elongations than materials obtained by traditional manufacture methods. This 3D printing process will enlighten a rational design of composites with complex shapes, and can also be extended to other thermoplastics based composites, including ABS, PC, and PA.

1. Introduction

For practical use, two or more materials are usually combined in order to rectify some shortcoming of a particularly useful component [1]. The overall composite will have better properties than the constituents. Because of their ease of processability, composite materials have been utilized to replacing conventional materials in many fields [2]. Particularly, emerging composites are widely used for advanced applications such as airframes, space structures, race cars, and sporting goods [3]. However, traditional fabrication methods for composite structures are very complicated due to the requirements of molds, forms, or dies, causing a limitation on the shape of the products and casting shadow on the wide applications of composites [4].

Three dimensional (3D) printing is an emerging technology, using computer aided design (CAD) data sets to produce 3D physical model, also known as additive manufacturing (AM) [5, 6]. The advantage of 3D printing lies in that products with complex shapes can be fabricated. After few decades of development, AM products have been used in a wide range of activities. The largest three rapid-increasing applications of AM fall into the areas of aerospace, automotive, and medical engineering. Other applications include architecture, education, fashion, and so on [7]. Among all the AM techniques, fused deposition modeling (FDM) is one of the most popular ones for thermoplastic parts because of its low material cost and user-friendly manipulation [8, 9]. This technology is mainly based on material extrusion through a nozzle [10]. Plastic filaments are constantly supplied

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into the heated nozzle. After melting, they are extruded through the nozzle and deposited on the platform where they immediately bond layer upon layer.

The materials commonly used in FDM are usually thermoplastic, including acrylonitrile butadiene styrene (ABS), polylactide (PLA), polycarbonate (PC), polyamide (PA), polyethylene, polypropylene and mixtures of them [11–13]. Since the intrinsic mechanical properties of the polymeric materials are relatively poor, a series of studies have been done to improve the properties by blending polymer with iron powders [14], carbon fibers [15, 16], nylon fibers [17], and so on. Nikzad *et al* investigated the thermo-mechanical properties of metal-polymer composites produced by FDM consisting of ABS containing 10% fine iron powders by volume [14]. They ground the ABS pellets to fine powders and then mixed them with iron powders in a ball-mill mixer followed by using a single screw extruder to fabricate the filaments. The results showed that the introduction of iron powders increased the storage modulus, glass transition temperature and thermal conductivity of the composites. A twin-screw extruder was used in the work of Perez *et al* to produce filaments of ABS-based materials, followed by 3D printing [18]. Jute fiber and TiO2 particle were added in different samples as reinforcing materials to compare the effects of different additives. To achieve more uniform mixing of metal and polymeric matrix, a twin-screw extruder and a single-screw extruder were both used in more studies [8, 19, 20]. The compatibility of reinforcing materials with matrix materials and the proportions of fillers can affect the mechanical properties of the obtained composite materials. Matsuzaki *et al* presented a method for fabrication of continuous fiber-reinforced thermoplastics based on FDM [21]. PLA filament was used as the matrix material. Carbon fibers and twisted jute natural plant fibers were used as continuous reinforcements. The tensile modulus and strength of the carbon fiber-reinforced PLA composites were greatly improved compared to that of pure PLA, whereas the improvement on tensile strength of jute fiber-reinforced PLA was not significant.

Although some efforts have been made to obtain reinforced plastic matrix composites with improved properties for FDM process, there are few reported studies on iron powders reinforced PLA. In research, it is necessary to use a large amount of raw materials. The disposal of materials can place a heavy burden on our environment. As environmental issues are becoming increasingly difficult to resolve, it is important for researchers to use green materials in experiments so that green products can be prepared for practical applications. Polylactic acid (PLA) is an environmentally friendly biodegradable polymer made from renewable agricultural raw materials [22]. Its basic monomer is lactic acid, which is derived from starch by fermentation [23, 24]. Mechanically, unoriented PLA is quite brittle, but possesses good strength, high modulus and stiffness [25, 26]. The brittleness can possibly be overcome by composing with other materials. Generally, PLA polymers are made into useful items using thermal processes, such as injection molding and thermoforming [27]. The shear viscosity of PLA decreases as the temperature increases, causing it flow more easily [28]. Therefore, PLA can be melt-processed at properly controlled temperatures in extruders, and the process of heating PLA in the nozzle of the 3D printer can also be achieved. Having a good understanding of thermal, crystallization, and rheological behaviors of the polymer is crucial for optimizing the process of its preparation and improving its mechanical properties.

The main reason for selecting iron powders as fillers is that they have good thermal and mechanical properties, and good interface interaction with polymers [14]. In addition, carbonyl iron powders are widely used as magnetic absorber in stealth technology [29]. Previously, many researchers chiefly concentrated on the development of low radar cross section (RCS) shapes and radar absorbing material (RAM) coatings [30]. Recently, investigation on the radar absorbing structures (RASs) using fiber reinforced polymeric composite materials has received more attention. RASs are multi-functional composites with lightweight and high strength, which can absorb electromagnetic wave and be used as load bearing parts at the same time [31, 32]. With 3D printing, the fabrication of complex RASs would be faster and easier than that with traditional methods.

This study aims to investigate the effect of 3D printing process on fabricating iron powders filled PLA composites, demonstrating its significant role in improving the performances of the as-obtained composites. Firstly, the technical process of preparing filaments for 3D printing and printing the final products by a 3D printer were conducted. The characterization of the morphologies and magnetic properties of the filaments were then carried out. The electromagnetic and mechanical properties of the final products were investigated using a broad band dielectric spectrometer and a universal testing machine, respectively. The complex permittivity and permeability of the samples relating to electromagnetic properties were analyzed. The tensile strength and elongation at break of the samples were also discussed in detail.

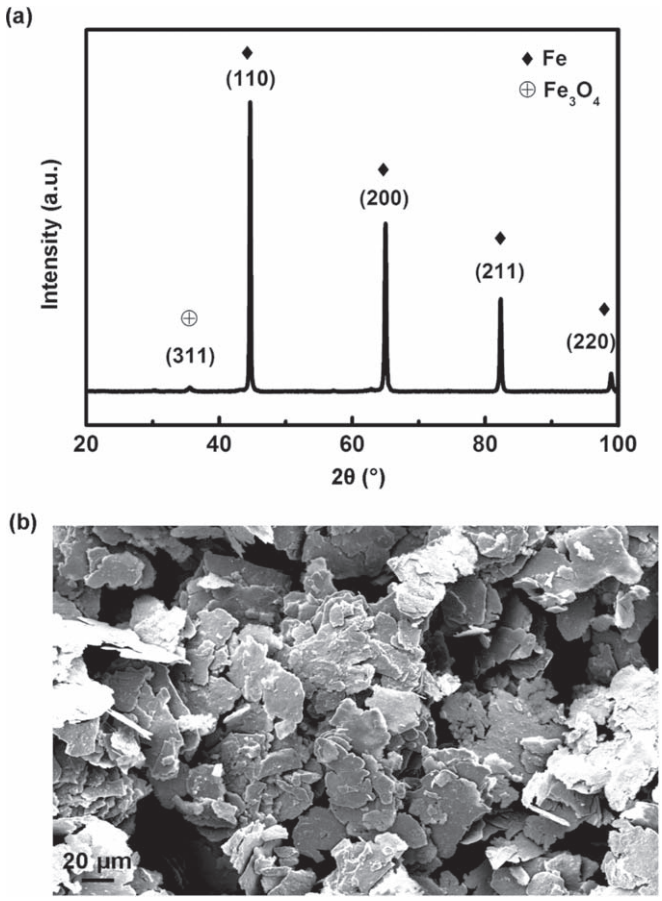


Figure 1. The characterization of carbonyl iron powders. (a) XRD pattern and (b) SEM morphology.

2. Experimental

2.1. Raw materials

The raw materials used in this study were carbonyl iron powders (Central Iron & Steel Institute, Beijing, China) and PLA pellets (2002D, NatureWorks LLC., Minnetonka, MN). The sizes of the carbonyl iron powders ranged from 2 μm to 160 μm according to a Malvern Instruments (Malvern, UK) particle size analyzer.

Figure 1(a) shows the XRD pattern of the carbonyl iron powders used in this study. The diffraction peaks at around $2\theta = 45°, 65°$, and 82° suggest that the Fe powders are of a body-centered cubic structure. The peak at $2\theta = 35.1°$ can be indexed to (311) reflection of Fe_3O_4, which demonstrates that the iron powders are partially oxidized. Scanning electronic microscopy (SEM) shows that the iron powders are flake shaped with different sizes and their surfaces are very smooth (figure 1(b)). Since flake-shaped iron powders are easier to be polarized and to achieve multiple reflection of incident electromagnetic waves, they present better wave absorbing capabilities than that with other shapes [33, 34]. It can be expected that the composites will demonstrate good electromagnetic properties.

2.2. Preparation of filaments

The filaments of the composites for 3D printing were prepared by a two-step mixture process. To make it clear, the temperature profiles of each step for different filaments are listed in tables 1 and 2. Firstly, the iron powders and PLA pellets were mixed in SHJ-20 twin-screw extruder (Haisi Extrusion Equipment Co., Ltd, Nanjing, China) with a screw rotation speed of 10 rpm. There were four temperature zones from the feeding section to the nozzle, as shown in figure 2(a). The temperature of the composite mixture at the nozzle was a very important parameter. If the temperature was too high, the mixture would not solidify after being extruded from the nozzle. Otherwise, if the temperature was too low, the viscosity would be too high for the materials to flow with ease. Thus, the temperatures of the four zones were set to be 150 °C, 165 °C, 170 °C, and 170 °C, according to the thermal properties of PLA. The raw materials were mixed in the extruder and extruded continuously. The obtained filaments were denoted as IPI-01, IPI-05, IPI-10, IPI-20, IPI-30, and IPI-50, and the weight percentages of carbonyl iron powders were 1%, 5%, 10%, 20%, 30%, and 50%, respectively. The filaments were

Table 1. First-step mixture using the twin-screw extruder.

Twin-screw extruder		
Raw materials	Temperatures of the four zones (°C)	Name of the product
1 wt% Fe + PLA	150, 165, 170, and 170 respectively	IPI-01
5 wt% Fe + PLA		IPI-05
10 wt% Fe + PLA		IPI-10
20 wt% Fe + PLA		IPI-20
30 wt% Fe + PLA		IPI-30
50 wt% Fe + PLA		IPI-50

Table 2. Second-step mixture using the single-screw extruder.

	Temperatures of the four zones (°C)	Name of the product
IPI-01	150, 165, 170, and 170, respectively	IPII-01
IPI-05		IPII-05
IPI-10		IPII-10
IPI-20	140, 150, 130, and 110, respectively	IPII-20
IPI-30		IPII-30
IPI-50		IPII-50

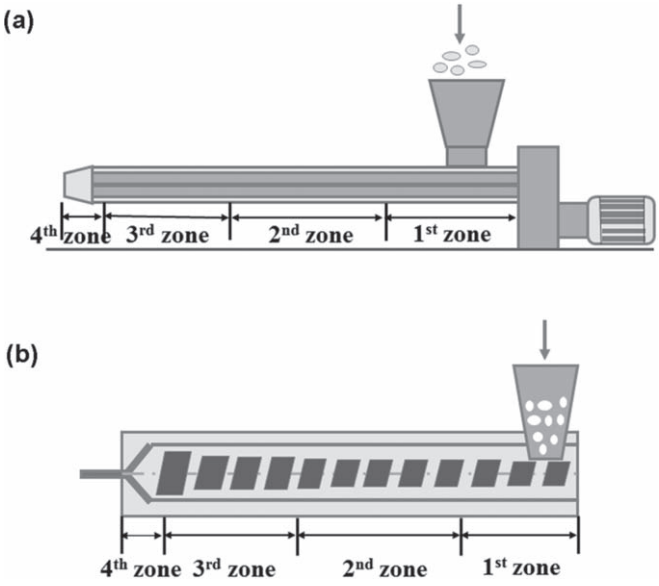


Figure 2. Diagrams of (a) the twin-screw extruder and (b) the single-screw extruder.

then shredded into small pieces by a mechanical chopper and dried in a vacuum oven at the temperature of 50 °C. The second-round mixing was carried out in Haake PolyLab single-screw extruder (Thermo Fisher Scientific Co., Ltd, Waltham, MA). The screw rotation speed was 20 rpm. The four temperature zones of Haake PolyLab single-screw extruder (figure 2(b)) were set to 150 °C, 165 °C, 170 °C, and 170 °C for IPI-01, IPI-05, and IPI-10. For IPI-20, IPI-30 and IPI-50, if the temperatures near the nozzle were too high, the materials been extruded would stay quasi-liquid state and could not be collected, so the temperature profile was set to 140 °C, 150 °C, 130 °C, and 110 °C. The diameter of the final filaments was controlled to be 1.75 mm by adjusting the extrusion speed and filament collecting speed. The as-prepared filaments were denoted as IPII-01, IPII-05, IPII-10, IPII-20, IPII-30 and IPII-50. After the two-step mixture process, we attained filaments that were suitable for

Table 3. Temperature profile of 3D printing.

	Temperature of the nozzle(°C)	Name of the product
IPII-01	190	IP01
IPII-05		IP05
IPII-10		IP10
IPII-20	185	IP20
IPII-30		IP30

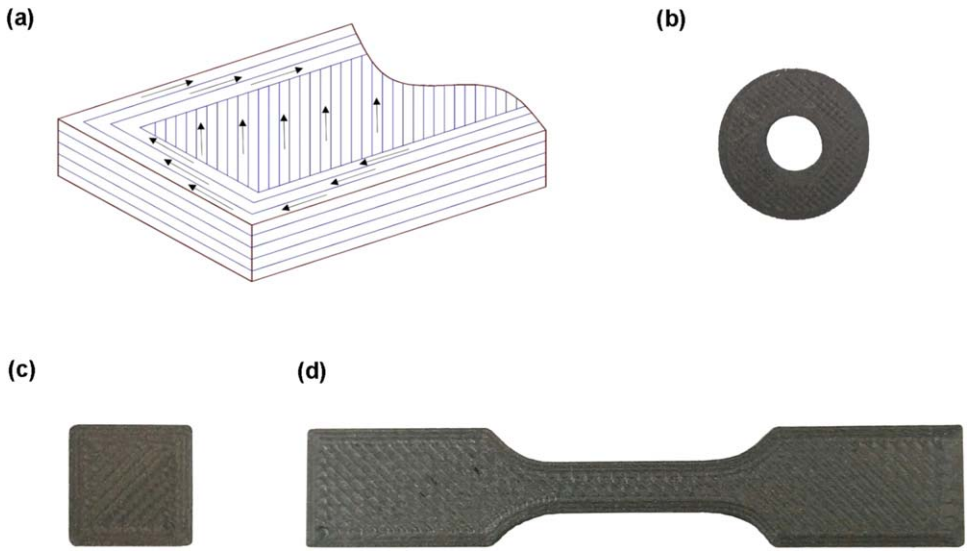


Figure 3. Illustration of 3D printing fabrication of the composites. (a) Schematic of the deposition process of the filaments from the printer head. The specimens for (b) complex permeability, (c) complex permittivity, and (d) mechanical tensile tests.

3D printing with a uniform distribution of iron powders. Pure PLA filaments were also fabricated by the same procedure as IPI-01 and IPII-01.

2.3. 3D printing process

A FDM 3D printer (AOD 3D Co., Ltd, Qingdao, China) was used to fabricate the dumbbell-shaped bars for tensile test, ring samples (inner diameter: 8 mm, outer diameter: 20 mm, thickness: 6 mm) and rectangular solid samples for measuring mechanical and electromagnetic properties. Since the IPII-50 filaments were too fragile for further manufacturing, they were not used for 3D printing. Pure PLA tensile bars were printed for comparison to measure the mechanical reinforcement of iron powders. Computer-aided-design (CAD) models were built first. The filaments were then supplied to the printer head. They were heated in the nozzle and extruded through the nozzle continuously. After several attempts, the appropriate extrusion temperatures for different materials were find out and listed in table 3. The temperature of the nozzle was set to 190 °C for pure PLA, IPII-01, IPII-05, and IPII-10 filaments. For IPII-20 and IPII-30 filaments, when the nozzle temperature was 190 °C, the viscosity was too high, so the melted filament cannot immediately bond on the platform. Thus, the temperature was set to 185 °C. The thickness of each layer was set to 0.1mm and the deposition directions (raster angles) for different layers were 45° and 135° alternatively (see figure 3(a)). The printing speed was 40 mm s^{-1}. The obtained samples by 3D printing were denoted as IP01, IP05, IP10, IP20 and IP30.

2.4. Characterization

The particle sizes and size distributions of the iron powders were determined by a particle size analyzer (Malvern Instruments, Malvern, UK). The morphology of the iron powders, filaments and final products was observed by field-emission scanning electron microscope (FESEM, Zeiss, Germany). The composite samples were sputter-coated with gold first because the PLA matrix is electrically non-conductive. Characterization of the iron powders was conducted using x-ray diffractometer (XRD, D/max-2500/PC, Rigaku Corporation, Tokyo, Japan). The 2θ angle range of the measurements was from 20 to 85°. Magnetic properties of the prepared filaments were tested by the vibrating sample magnetometer (VSM, Lake Shore Cryotronics, Inc., CA). A broad band dielectric spectrometer (Concept-80, NOVOCONTROL Technologies GmbH & Co. KG, Germany) was

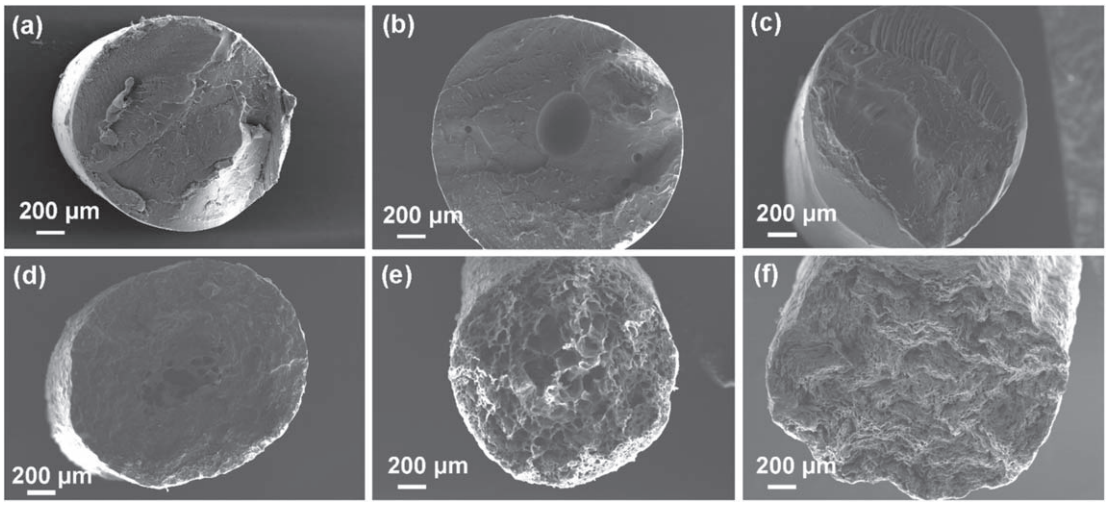


Figure 4. SEM images of (a) IPI-01, (b) IPI-05, (c) IPI-10, (d) IPI-20, (e) IPI-30, and (f) IPI-50 filaments.

used for measuring the electromagnetic property of the composite samples. The main component was Agilent E4991A RF impedance/material analyzer. The ring and rectangular samples were fixed in the electromagnetic shielded holders. The frequency range was set to 10 MHz–2 GHz. The data of real and imaginary permittivity and permeability were collected by computer. The test proceeded under room temperature. The fabricated tensile bars dimensioned in ASTM D638 standard were tested using a universal testing machine (Shanghai Jiubin Instrument Co. Ltd, China). The tensile test employed two grips with one fixed and the other movable to hold the specimens. The data of force (N) and displacement (cm) were collected by computer. The testing speed was 5 mm min^{-1}. For each kind of composite samples, at least four specimens were tested.

3. Results and discussions

3.1. Filaments

The filaments with different contents of iron powders were prepared through a two-step mixture of the raw materials. The extrusion temperatures were elaborately adjusted to avoid clogging of the mixtures in the chamber. The two-step mixing process enabled the iron powders to be evenly dispersed in the PLA matrix.

The morphologies of the cross-sections of the filaments after first-step mixture were identified by SEM. The results are shown in figure (4). Actually, the iron powders we used were flaky, meaning they were very thin. The high aspect ratio fillers would align along the extrusion direction under the shear within the nozzle when extruded from the extruder. Figure (4) shows the cross-section of the filaments, so we should have seen the lateral side of the fillers. However, since the surfaces of the cross-section are very coarse (affects the contrast of SEM image) and the lateral dimension of the fillers is very thin, we can hardly see the fillers.

It can be observed that the porosity of the filaments becomes higher as the content of iron powders increases. When the weight percentage of fillers is low (1%, 5% and 10%), the cross-sections are very smooth. When the weight percentage of fillers is 20%, many gas evacuated pores generated by extrusion appear on the surface, especially in the central area. On the cross-section of IPI-30 sample, the density of the pores distribution is even higher that makes the filament look like a sponge. Additionally, the sizes of the pores in the central area are greater than that of the pores located on the periphery of the filament. The reason may be the temperature difference between the central area and the periphery. The pressure from the twin screws can also affect the pore size. When the content of iron powders is 50% and thus the weight percentage of the PLA matrix is much lower, there are too many pores in the material with the result that the filament cannot maintain its round shape (figure 4(f)). The difficulty of the densification of iron powders can be one reason for this result. In addition, in the process of single-screw extrusion, the temperatures of each temperature zone for IPI-20, IPI-30 and IPI-50 were lower than that of IPI-01, IPI-05 and IPI-10. The decrease in temperature that left the composites with low viscosity may also result in higher porosity of the filaments [35].

Figure 5 shows the room temperature magnetic hysteresis loops of the filaments measured in the magnetic field range of −10 000–10 000 Oe. The insets in figures 5(a) and (b) show the magnified images of the loops. The hysteresis loops of all samples are of an S-like shape, and the remanence (M_r), coercive force (H_c) and hysteresis losses are relatively low, indicating a typical soft ferromagnetism behavior [36]. Thus, they are suitable for applications in dynamos, alternating electromagnetic magnets and transformers. The different magnetic

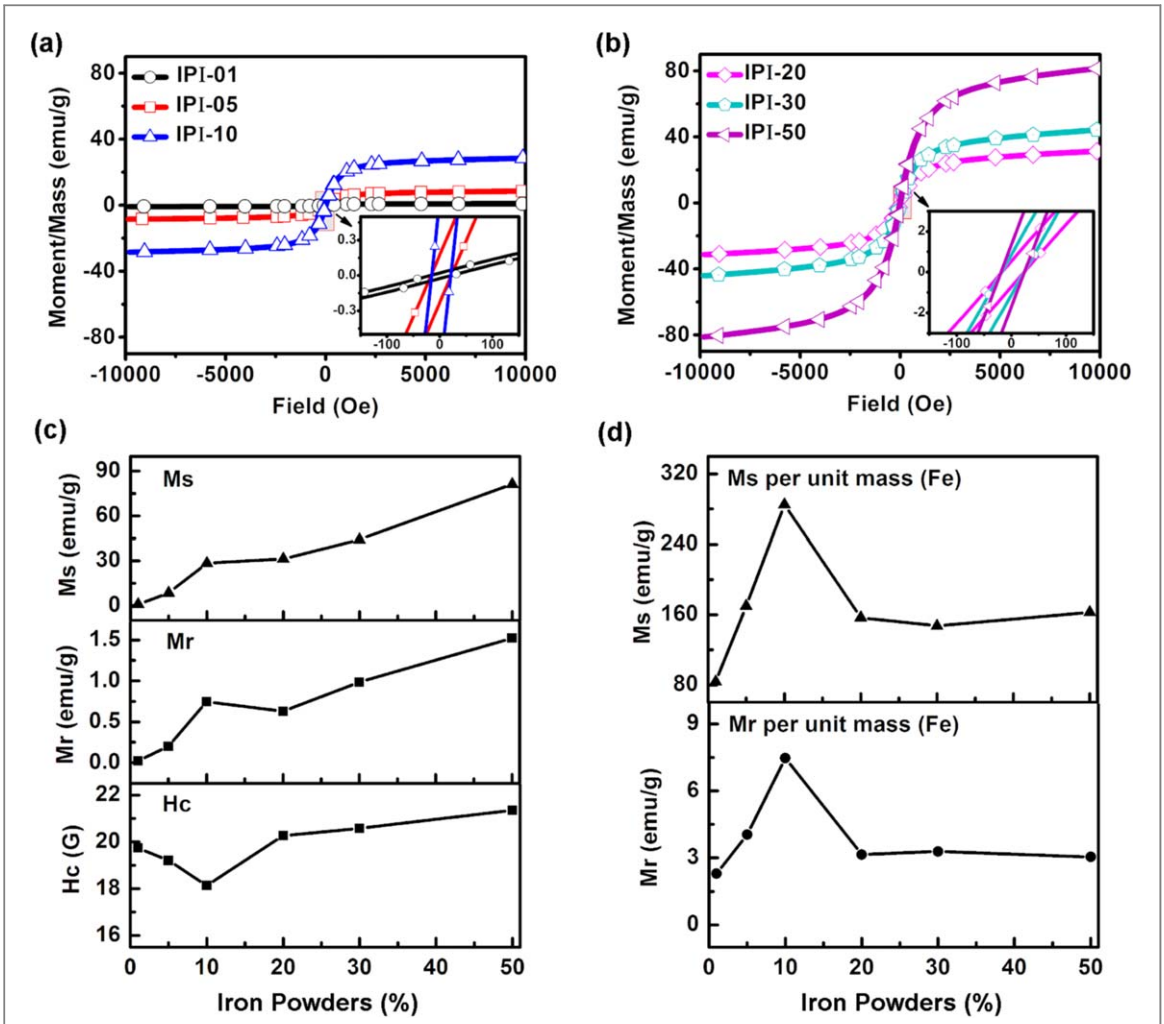


Figure 5. Magnetic properties of the filaments: the M-H loops of the filaments (a) IPI-01, IPI-05, and IPI-10, (b) IPI-20, IPI-30, and IPI-50, (c) M_s, H_c, and M_r values of the filaments, and (d) M_s and M_r values per unit mass of Fe of the filaments.

Table 4. Magnetic parameters of the composite filaments.

Iron powders contents/%	M_s/emu/g	M_i^s/emu/g(Fe)	M_r/emu/g	M_i^r/emu/g(Fe)	H_c/ Oe
1	0.836	83.60	0.023	2.300	19.741
5	8.493	169.86	0.202	4.040	19.207
10	28.475	284.75	0.747	7.470	18.141
20	31.322	156.61	0.630	3.150	20.267
30	44.137	147.12	0.986	3.287	20.584
50	81.282	162.56	1.523	3.046	21.354

parameters of the samples are listed in table 4. From figures 5(a) and (b), it can be seen that the saturation magnetization (M_s) and M_r values of the samples increase with increasing the iron powders content. When the fillers content is 1%, the M_s value is 0.836 emu g^{-1}; and the M_s value increases to 81.282 emu g^{-1} when the fillers content is 50%. The M_r values change from 0.023 to 1.523 emu g^{-1} when the iron powders content increases from 1% to 50%.

Although the overall variations of M_s and M_r increase with increasing the fillers content, it can be noticed in figure 5(c) that the relationship between the values of M_s and M_r and the iron powders content is not linear. To probe deeper into this outcome, the saturation magnetization (remanence) per unit mass of iron powders M_i^s (M_i^r) is calculated, and the results are shown in table 4 and figure 5(d). It is found that the peak values of both M_i^s and M_i^r appear at IPI-10 sample, which indicates there could be an optimum value of iron powders content for the composites. Although M_s and M_r of IPI-20, IPI-30 and IPI-50 samples are greater than that of IPI-10 sample, their values of M_i^s and M_i^r are less than that of IPI-10 sample. This result can be attributed to the interaction among the carbonyl iron particles. When the fillers content is low, the iron powders are dispersed and immersed

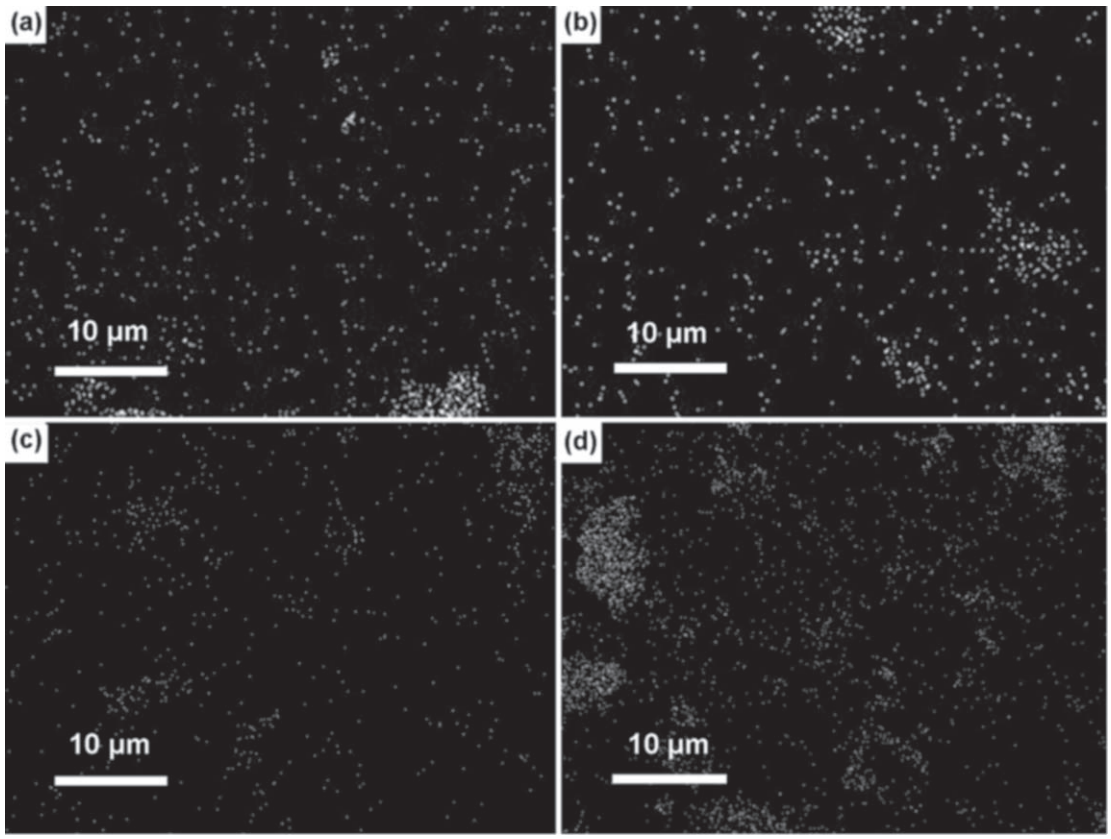


Figure 6. Iron powders distributions in the cross-section of (a) IP05, (b) IP10, (c) IP20, and (d) IP30 samples.

in the PLA matrix separately, so the interaction between iron particles is relatively weak. When the content increases, the gaps between iron particles becomes smaller, and the interaction between adjacent iron particles becomes stronger. However, when the fillers content is too high, some iron particles will stick together so that some particles may contribute less to M_s and M_r The result shows that when the fillers content is 10%, the interaction among carbonyl iron particles is the best for the magnetic behavior of the composite.

As can be seen in figure 5(c), H_c values decrease at first and then increase with increasing iron powders content. When the iron powders content is 1%, H_c value is 19.741 Oe. As the fillers content increases to 10%, H_c value of the filament decreases to 18.141 Oe, which is the least one among all the composites. The H_c value then increases again with the increase of the weight percentage of iron powders. The greatest value of H_c is 21.354 Oe when the fillers content is 50%. This result can be attributed to the microscopic structure of the composites, which varies from solid to porous with the increase of iron powders content. Higher porosity may result in higher internal stress. Moreover, when the contents of the components are different, the internal stresses of the composites are also different.

When the carbonyl iron powders content is 10%, the filaments exhibit the greatest saturation magnetization and remanence per unit mass of Fe, and the minimum value of coercive force. Therefore, it can be concluded that the iron powders filled PLA composites exhibit the best magnetic properties when the fillers content is 10%.

3.2. Electromagnetic properties

To investigate the electromagnetic properties of the composite materials, ring-shaped and rectangular solid samples were manufactured by FDM using the as-prepared filaments. To examine the distribution of iron powders in PLA matrix, EDS analysis was used. The results are shown in figure 6. In the pictures, the bright dots represent carbonyl iron powders and the black background represents PLA matrix. Iron powders are uniformly distributed in the composites with lower iron powders contents (figures 6(a), (b)), and particle aggregations randomly exist in some regions. When iron powders contents are higher, particle aggregation is more likely to happen because the distribution density of iron particles increased (figures 6(c), (d)). The differences in the distribution can be ascribed to the differences in the flow properties of the melted mixtures. Mixtures with higher contents of iron powders have higher viscosities, resulting in bad dispersion of the particles that leads to aggregation [37]. In general, the iron powders present homogeneous dispersion in the samples. The results support that the two-step mixture process can achieve a good mixing effect.

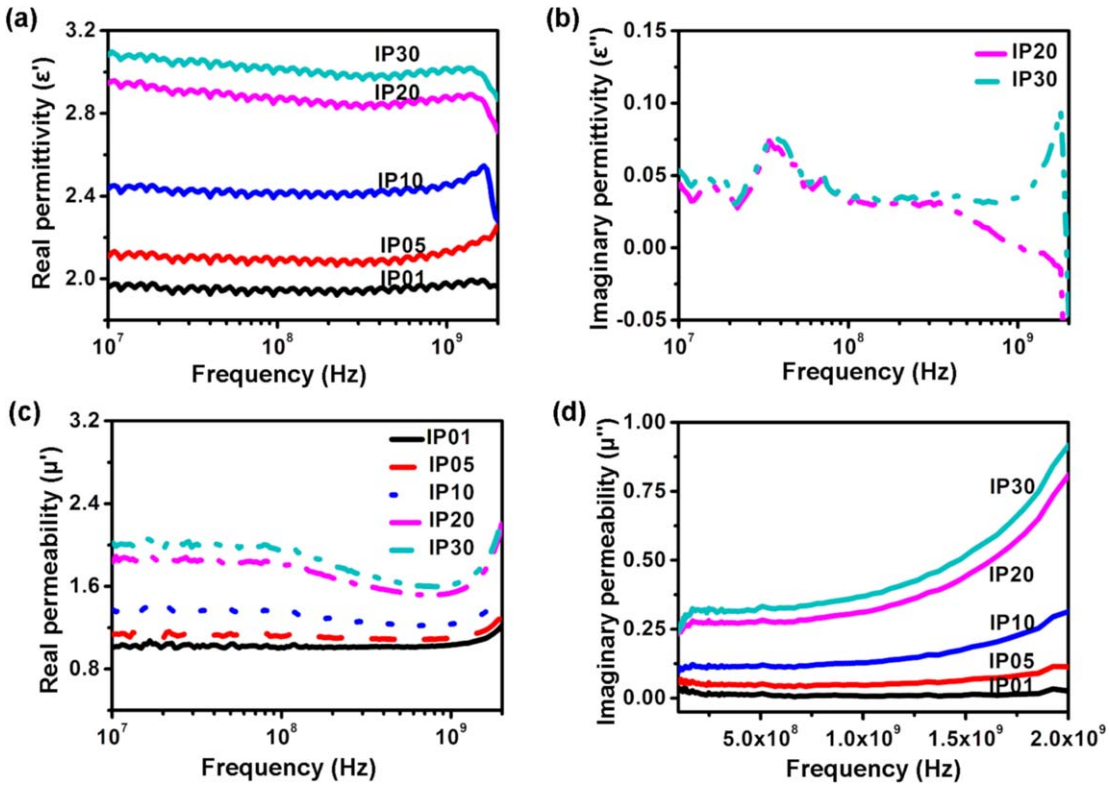


Figure 7. Electromagnetic properties. (a) Real and (b) imaginary permittivity of the rectangular solid samples. (c) Real and (d) imaginary permeability of the ring samples.

The complex permittivities and permeabilities of the samples are shown in figure 7. For IP01, IP05 and IP10 samples, their dielectric loss angles are too small to be precisely measured and the tangent values of the dielectric loss angles include negative values, so the imaginary permittivity of these samples are not plotted in figure 7(b). It is known that the real parts of complex permittivity (ε') and permeability (μ') represent the storage ability of electromagnetic energy, while imaginary parts (ε'' and μ'') indicate the inner dissipation [38]. As the content of iron powders increases, the values of both ε' and ε'' become higher, implying the storage capability and dielectric loss increase with the increase of filler content. However, the values of ε' of IP01, IP05, and IP10 are almost independent of frequency, while for IP20 and IP30 samples ε' shows a slow decline as the frequency increases. The peaks appear in ε'' of IP20 and IP30 should be ascribed to the polarization related relaxations and resonance [39]. The peaks are located between 10 MHz and 100 MHz, indicating a better performance over this frequency range. The complex permittivity values of IP01, IP05, IP10, IP20 and IP30 samples are 1.95, 2.11, 2.44, 3.00, and 2.88, respectively. Overall, the complex permittivity values of the samples fall into 2–3, indicating that the dielectric loss of the composites are low.

At the same value of frequency, the values of real and imaginary permeability increase with increasing the content of iron powders (figures 7(c), (d)). There exists a huge increase for both μ' and μ'' especially when the content changes from 10% to 20%. The slight increase from IP20 to IP30 indicates that when the filler content exceeding 20%, the introduction of more fillers contributes less to the magnetic storage ability. For a given sample, the values of μ' are almost stable when the frequency is between 10 and 100 MHz. In the frequency of ranging from 100 MHz to 2 GHz, the values of μ' of IP01 and IP05 samples increase moderately with the increase of the frequency; while for IP10, IP20 and IP30 samples, the values of μ' first decrease and then increase with the increase of the frequency. Moreover, this decrease is more and more severe as the content of iron powders increases, which needs further explanations. One possible reason is that the aggregation of the iron particles hinders the electromagnetic absorption properties of the composites, but why it only happens at high frequency is still to be studied. It is indicated that the composite materials have a relatively worse capacity of propagation and storage of magnetic field in low frequency bands compared with that in high frequency bands. The imaginary permeability (μ'') indicates the magnetic loss. The value of μ'' increases with the increase of iron filler. This can be explained by the equation $\mu'' \propto M_s^2/K_1$, where K_1 is the magnetocrystalline anisotropy constant and the saturation magnetization (M_s) values [40]. Since K_1 is the same for different samples, the value of μ'' changes along with the variation of saturation magnetization. The values of μ'' also continually increase with the increase

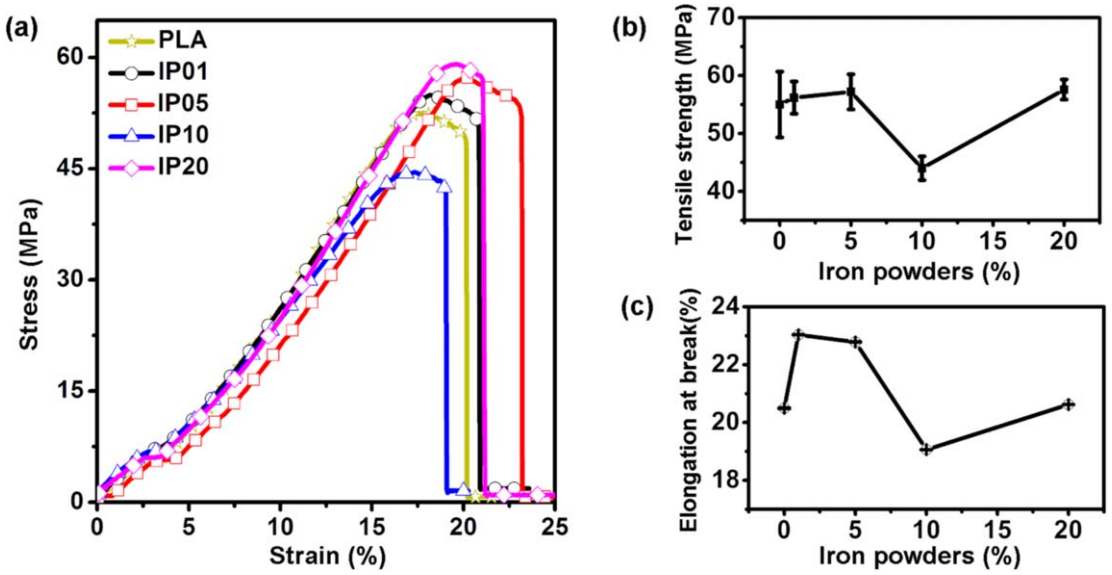


Figure 8. Tensile properties of the samples. (a) Representative stress–strain curves of the samples with different iron contents. (b) Tensile strength. (c) Elongation at break.

of frequency value in the range of 100 MHz–2 GHz, showing that the composite materials have a better wave absorbing property in high frequency bands.

The 3D printing process retained the electromagnetic properties of the iron powders, so the composite materials we fabricated have a better magnetic lossy property than dielectric lossy property. It can be expected to further improve the electromagnetic properties by adding dielectric lossy materials into the matrix, such as carbon fibers.

3.3. Mechanical properties

Tensile samples were prepared by FDM using the filaments to test the mechanical properties of the composite materials. Since the IPII-50 filaments were too fragile for 3D printing and the printed IP30 broke very easily, we could not collect the mechanical data of them. Figure 8(a) plots the stress as the function of strain of the tested tensile samples. The stress and strain have a linear relationship at the first stage where Hooke's Law holds. The samples are then deformed by plastic deformation until fracture.

The tensile strength and elongation at break of pure PLA products fabricated by traditional manufacturing processes are 40–60 MPa and 4%–10%, respectively [41, 42], whereas the values of our 3D printed pure PLA are 55 ± 5.68 MPa and 20.5 ± 0.024%, respectively. It means that 3D printing can substantially improve the plasticity of PLA while maintaining its strength. The stratified structure of the 3D printed samples may have enhanced the deformation capability of the polymer. The phenomenon may also include the effect of interactions between adjacent layers associated with 3D deposition process. Moreover, it is found that the raster angle can significantly affect the mechanical properties of FDM components [43]. It is very likely that the raster angles of 45° and 135° alternatively for different layers may enhance the ductility.

The printed composites show a slight increase in tensile strength compared with the reference samples produced using neat PLA except IP10 samples (figure 8(b)). The average tensile strength values of the samples are 56 ± 2.79, 57 ± 3.04, and 58 ± 1.74 MPa for IP01, IP05, and IP20, respectively. Furthermore, the printed composites demonstrate somewhat higher ductility (except IP10) as their elongations at break exhibit an increase over neat PLA from 20.5 ± 0.024% to 23.0 ± 0.012%, 22.8 ± 0.018%, and 20.6 ± 0.006% for IP01, IP05, and IP20, respectively (figure 8(c)). However, some previous work on 3D printed polymer-metal hybrid materials obtained different results [44, 45]. It was found that the tensile strength decreased with increasing the loading of metal particles, and increased temperatures result in increased elongation at break. In our study, temperature was also a variable. Thus, whether the incorporation of iron powders as fillers resulted in the improvement of mechanical properties still needs further investigation. Anyhow, it was found that better filler/matrix adhesion could increase the ductility of the composite material and lead to larger mechanical property increases [46], meaning there is a good interfacial adhesion between iron powders and the matrix in the samples we fabricated. When a load is applied, the matrix material yields first and passes the force to fillers, which will absorb plastic deformation energy and bear some of the applied load. Thereby the stress is more evenly

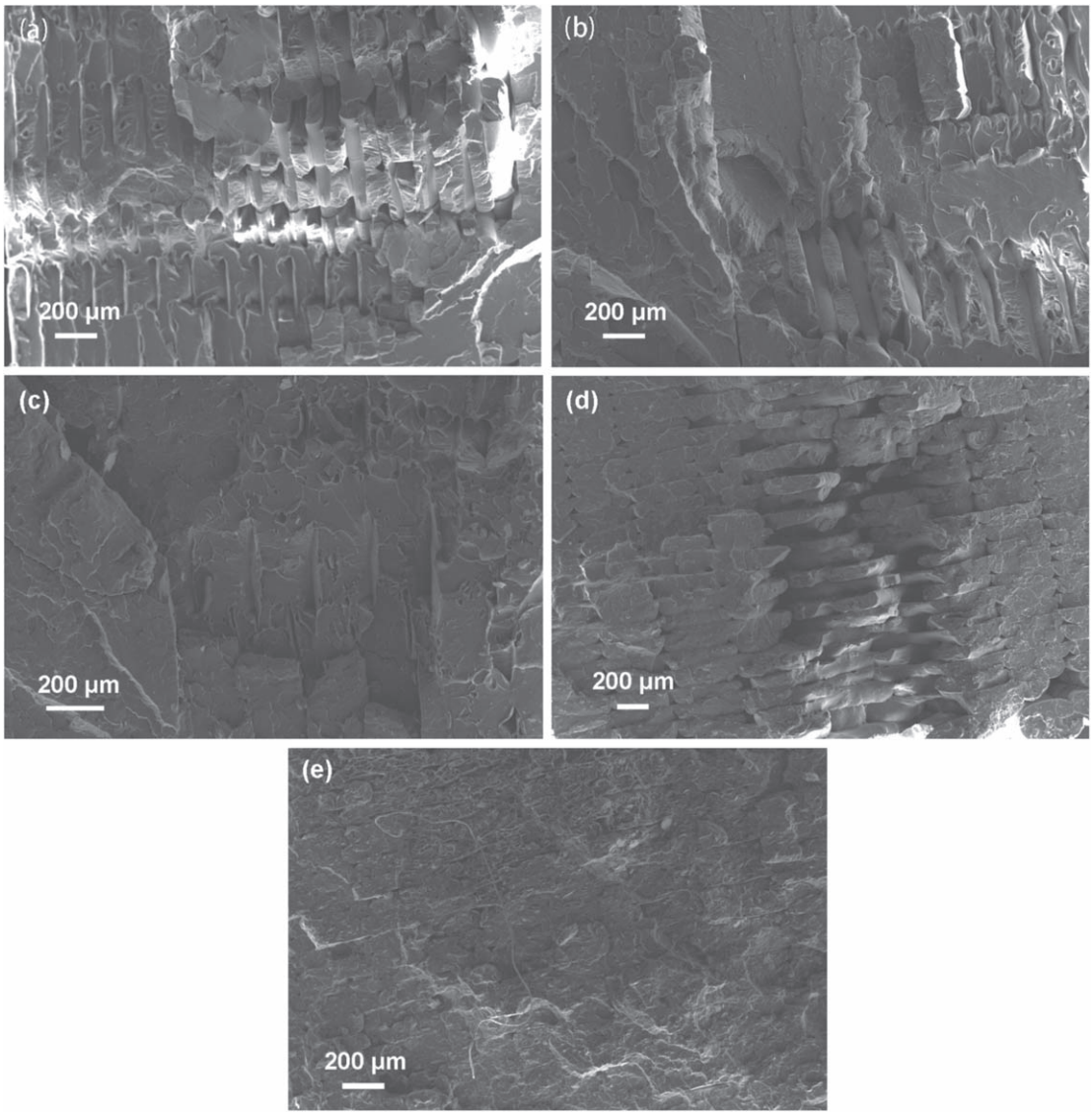


Figure 9. Fracture surfaces of (a) PLA, (b) IP01, (c) IP05, (d) IP10, and (e) IP20 samples.

distributed. Moreover, in the process of deformation, dislocations will pile up and tangle around PLA-iron particle interface so that to lower the velocity of crack propagation.

Tensile fracture surfaces were analyzed by SEM. The results provide more information about the fracture mechanism of the composites materials (figure 9). No deposition-related defects like filament clusters can be seen in the fracture surfaces. In addition, boundaries between different layers were hardly observed, indicating the excellent coalescence of the fiber-like materials extruded from the nozzle. It can be observed that there are many sags and crests in the tensile fracture cross-section of IP01, IP05, and IP20 samples (figures 9(b), (c), (e)), which is the feature of ductile fracture; while the morphology of IP10 samples is relatively smooth (figure 9(d)), which suggests the tensile fracture is brittle break. The different fracture modes partly explains the reduction of IP10 samples on the tensile strength and elongation at break. We can also see many large voids in the fracture surface of IP10 samples. The voids are mainly formed during the deposition process layer by layer, which indicates the interior adhesion of different layers are not good. Since the voids can result in stress concentration, they can cause the samples to break at lower stresses. Moreover, the porous structure of IP20 samples could be the cause of its improvement on tensile strength and the abatement on elongation at break, since the texture causes the mechanical properties of each printed layer [47].

4. Conclusions

In this paper, a novel process of fabricating carbonyl iron powders filled PLA composite materials was considered for improving the synergistic effect between the filler and the matrix. The parameters we paid major

attention to were: (a) the dispersion of the iron powders in PLA matrix, (b) the magnetic properties of the composites, (c) the complex permittivity and permeability of the composites, and (d) the mechanical properties and the way of fracture of the composites and pure PLA. The morphology of the filaments and fracture surface of the tensile samples were observed through SEM. Energy dispersive spectroscopy analysis was used to examine the distribution of iron powders. Vibrating sample magnetometer analysis was used to determine the magnetic properties. Electromagnetic and tensile tests were carried out on a broad band dielectric spectrometer and a universal testing machine, respectively. The following conclusions are drawn from the main results of this paper:

(1) The two-step mixture process enabled good dispersion of iron powders in the PLA, achieving effective interaction between the filler and the matrix.

(2) The 3D printing process maintained the magnetic properties of carbonyl iron powders. The saturation magnetization and remanence of the composites exhibited increases with the increment of carbonyl iron powders content. The composite with the fillers content of 10% by weight exhibited the largest M_s and M_r values per unit mass of Fe, which indicated that the optimum magnetic state of the composites was that with carbonyl iron powders content of 10%.

(3) The electromagnetic properties of the composites increased with the increase of carbonyl iron powders content. The complex permittivity values of the samples fall into 2–3, indicating the dielectric loss of the composites are low. The imaginary permeability increases with the increase of frequency value in the range of 100 MHz–2 GHz, showing that the composite materials have better wave absorbing property in high frequency bands.

(4) As for mechanical properties, though the tensile strength of our 3D printed pure PLA (55 $\pm$ 5.68 MPa) was comparable to traditional manufacturing (40–60 MPa), the elongation at break increased significantly (from 4%–10% to 20.5 $\pm$ 0.024%). The average tensile strength values of the composites are 56 $\pm$ 2.79, 57 $\pm$ 3.04, and 58 $\pm$ 1.74 MPa for IP01, IP05, and IP20 samples, respectively. And the printed composites demonstrate somewhat higher ductility (except IP10) as their elongations at break exhibit an increase over 3D printed neat PLA from 20.5 $\pm$ 0.024% to 23.0 $\pm$ 0.012%, 22.8 $\pm$ 0.018%, and 20.6 $\pm$ 0.006% for IP01, IP05, and IP20, respectively. Noticeably, while IP10 possessed the greatest magnetic properties, its mechanical properties were the worst, requiring further explanations.

Results support the feasibility of applying FDM based 3D printing process for composites with different shapes. Armed with this success, we can fabricate other thermoplastics based composite materials for different use. For example, to obtain a material with excellent wave absorbing and mechanical properties, adding dielectric lossy materials and reinforcing materials like carbon fibers would be a feasible choice. Furthermore, we can still improve the mechanical properties by optimizing the printing process.

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Conflict of interest

The authors declare no conflict of interest.

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